

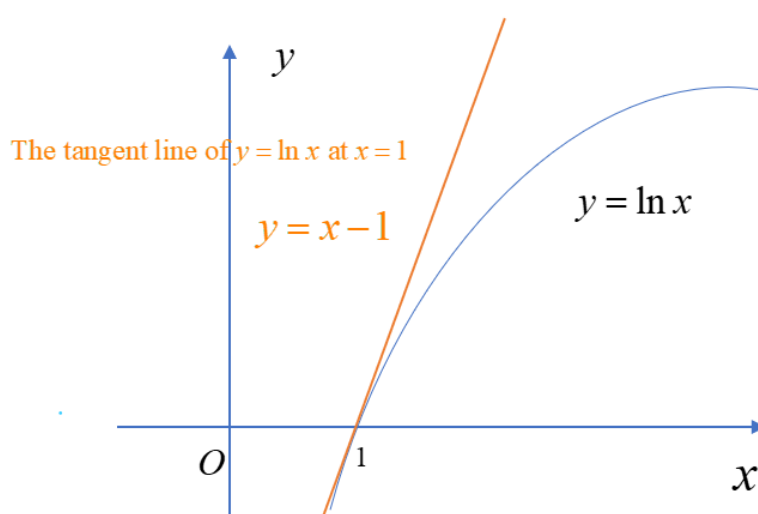
3.9 Linearization and Differentials

It is often useful to approximate complicated functions with simpler ones that give the accuracy we want for specific applications and at the same time are easier to work with than the original functions. The approximating functions discussed in this section are called *linearization*, and they are based on tangent lines. For example,

$$\ln 1 = 0,$$

$$\ln 1.001 = ?$$

But we check the following graph, we see that



when x is close to 1, the value of $y = \ln x$ is very close to the value of $y = x - 1$. That is

$$\ln x \approx x - 1, \quad |x - 1| \rightarrow 0.$$

So,

$$\ln 1.001 = 1.001 - 1 = 0.001.$$

The approximating function $y = x - 1$ is called linearization of $y = \ln x$ at $x = 1$.

① Linearization

As you can see in the above Figure, the tangent to the curve $y = \ln x$ lies close to the curve near the point of $x = 1$. For a brief interval to either side, the y -values along the tangent line give good approximations to the y -values on the curve. We observe this phenomenon by zooming in on the two graphs at the point of $x = 1$. The phenomenon

is true not just for the curve $y = \ln x$; every differentiable curve behaves locally like its tangent line.

DEFINITIONS If f is differentiable at $x = a$, then the approximating function

$$L(x) = f(a) + f'(a)(x - a)$$

is the **linearization** of f at a . The approximation

$$f(x) \approx L(x)$$

of f by L is the **standard linear approximation** of f at a . The point $x = a$ is the **center** of the approximation.

Example 1 Find the linearization of $f(x) = \sqrt{1+x}$ at $x = 0$.

Note: In Example 1, $\sqrt{1+x} \approx L(x) = 1 + \frac{1}{2}x$, if $|x - 0| = |x|$ is very small. So

$$\sqrt{1.2} = \sqrt{1+0.2} \approx 1 + \frac{1}{2}(0.2) = 1.1;$$

$$\sqrt{1.05} = \sqrt{1+0.05} \approx 1 + \frac{1}{2}(0.05) = 1.025;$$

$$\sqrt{1.005} = \sqrt{1+0.005} \approx 1 + \frac{1}{2}(0.005) = 1.0025.$$

Approximation	True value	True value – approximation
$\sqrt{1.005} \approx 1 + \frac{0.005}{2} = 1.00250$	1.002497	$0.000003 < 10^{-5}$
$\sqrt{1.05} \approx 1 + \frac{0.05}{2} = 1.025$	1.024695	$0.000305 < 10^{-3}$
$\sqrt{1.2} \approx 1 + \frac{0.2}{2} = 1.10$	1.095445	$0.004555 < 10^{-2}$

Example 2 Find the linearization of $f(x) = \sqrt{1+x}$ at $x = 3$.

Example 3 Find the linearization of $f(x) = \cos x$ at $x = \frac{\pi}{2}$.

② Differentials

We sometimes use the Leibniz notation dy/dx to represent the derivative of y with respect to x . Contrary to its appearance, it is not a ratio. We introduce new variables dx and dy , called *differentials*, and define them in a way that makes Leibniz's notation for the derivative dy/dx a true ratio.

DEFINITION Let $y = f(x)$ be a differentiable function. The **differential dx** is an independent variable. The **differential dy** is

$$dy = f'(x)dx.$$

Example 4

(a) Find dy if $y = x^5 + 37x$.

(b) Find the value of dy when $x = 1$ and $dx = 0.2$.

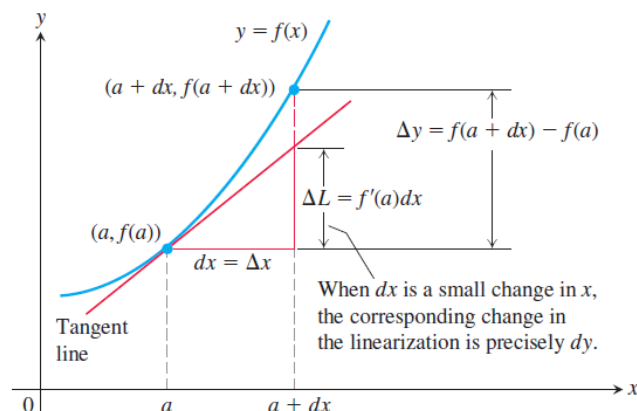
③ The Geometric Meaning of Differentials

The geometric meaning of differentials is shown as followings. Let $x = a$ and set $dx = \Delta x$. The corresponding change in $y = f(x)$ is

$$\Delta y = f(a + dx) - f(a).$$

The corresponding change in the linearization L is

$$\Delta L = L(a + dx) - L(a) = \underbrace{f(a) + f'(a)[(a + dx) - a]}_{L(a + dx)} - \underbrace{f(a)}_{L(a)} = f'(a)dx.$$



That is, the change in the linearization of f is precisely the value of the differential dy when $x = a$ and $dx = \Delta x$. Therefore, dy represents the amount the tangent line rises or falls when x changes by an amount $dx = \Delta x$.

Since $f(a) = L(a)$ and $f(x) \approx L(x)$ if $|x - a|$ is very small, we see that

$$\Delta y \approx \Delta L = f'(a)dx = dy$$

if $|x - a|$ is very small. So, we have the followings.

Note The differential dy is the main part of the corresponding change Δy when $dx = \Delta x$.

If $dx \neq 0$, then the quotient of the differential dy by the differential dx is equal to the derivative $f'(x)$ because

$$dy \div dx = \frac{f'(x)dx}{dx} = f'(x) = \frac{dy}{dx}.$$

Example 5 Find differentials of functions.

(a) $d(\tan 2x);$

(b) $d\left(\frac{x}{x+1}\right).$